

Figure 1 — Operational Performance Indicator (OPI) conceptual schematic

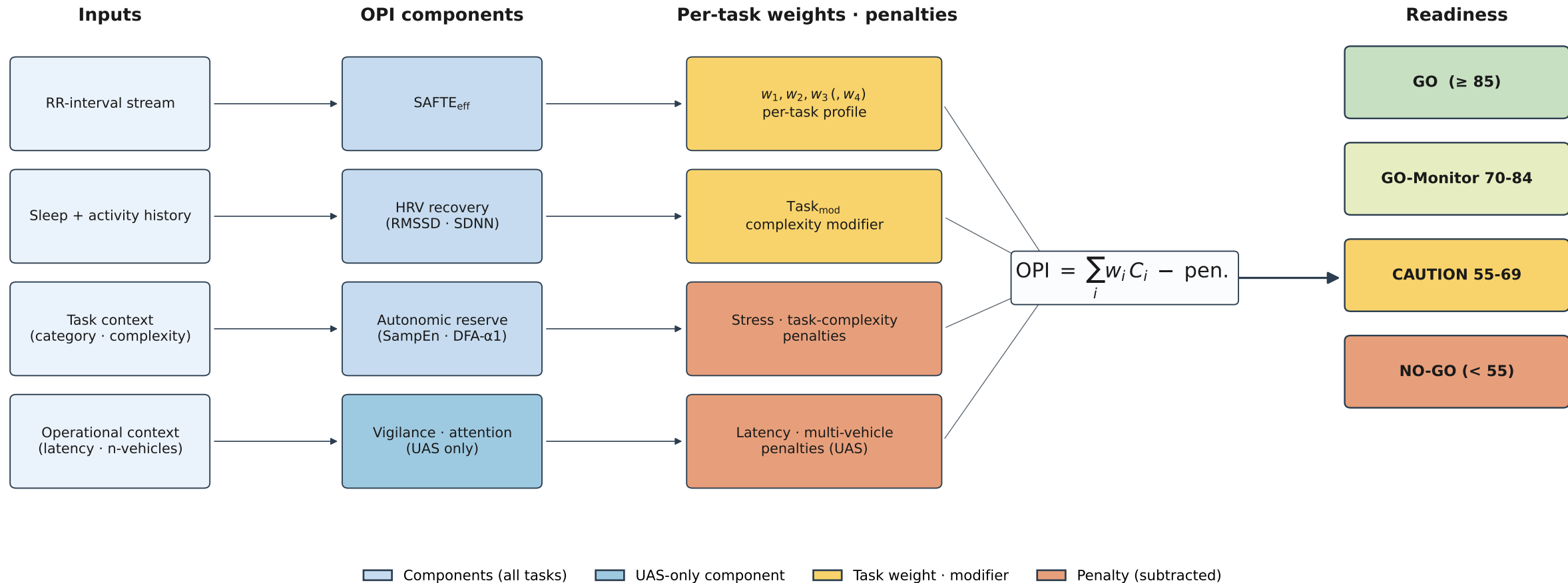
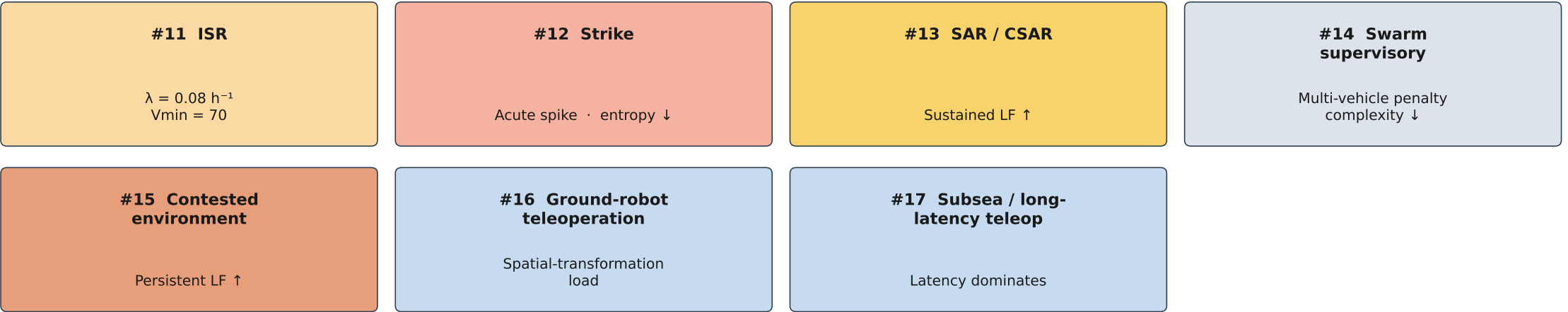


Figure 2 — Task taxonomy and dominant autonomic signatures

A. Manned aviation task categories (n = 10)



B. UAS / teleoperator task categories (n = 7)



LF/HF ↑ dominant Vagal withdrawal Complexity / entropy ↓ Precision-motor markers Acute stress suppression Vigilance decay (UAS) Latency-dominant (UAS)

Figure 3 — Illustrative OPI worked example: same HRV input, three task hypotheses

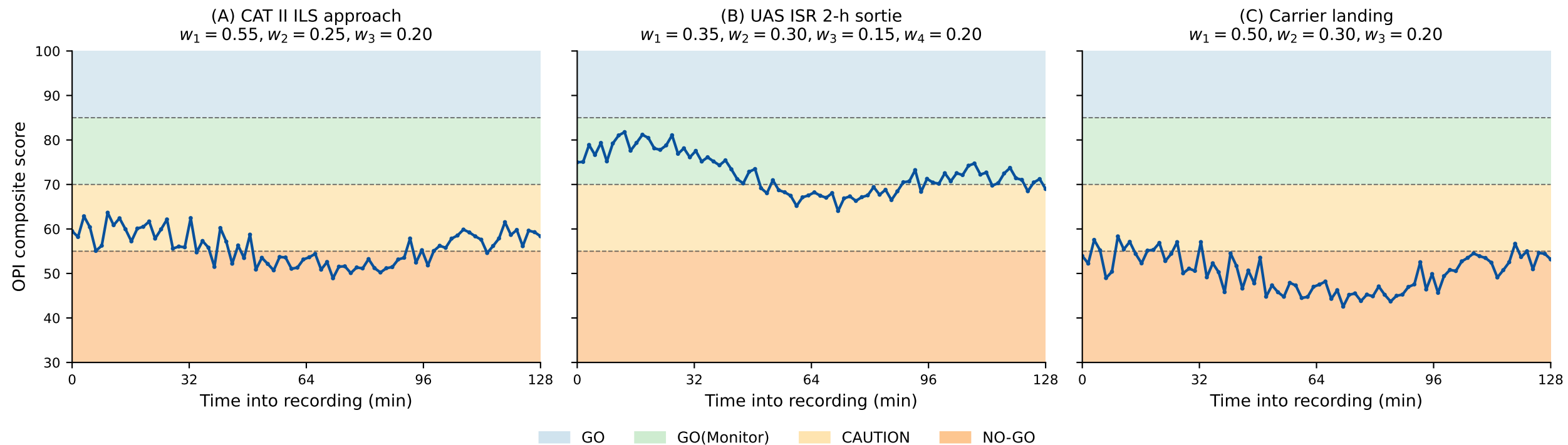
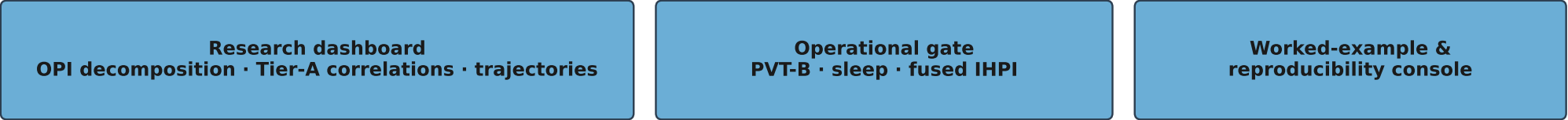
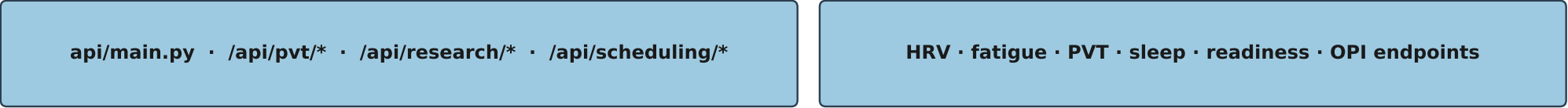


Figure 4 — Reference-implementation architecture

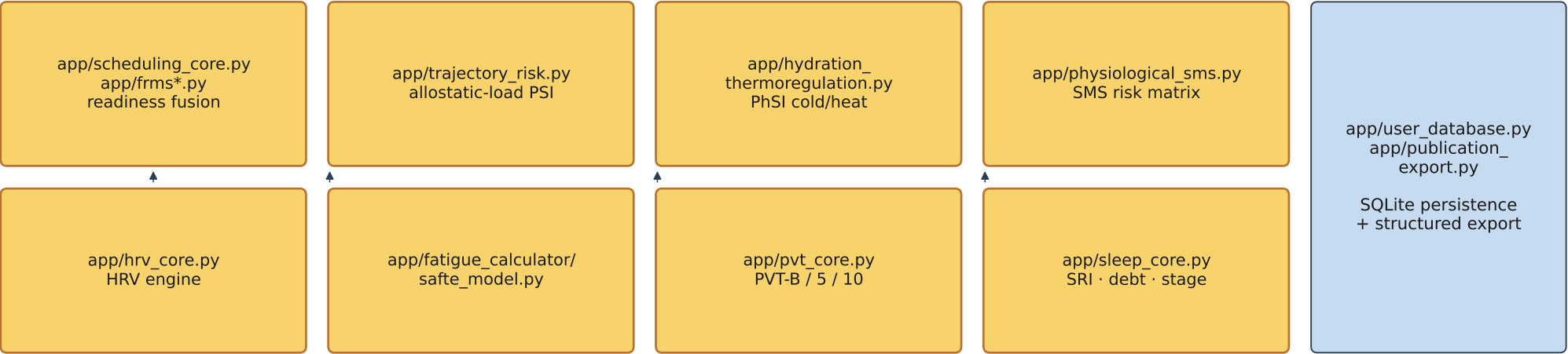
L4 — Next.js



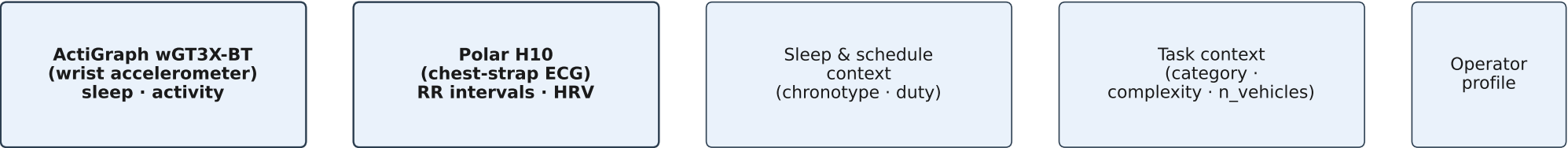
L3 — FastAPI



L2 — Python core



L1 — Inputs



Gold-bordered modules compose the OPI fusion pathway. Antarctic validation instruments boxed in bold at L1.